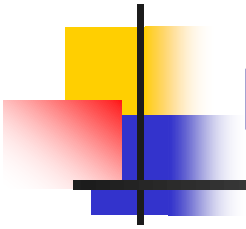


Pumping Lemma

Reading: Chapter 4

Some languages are *not* regular



When is a language is regular?

if we are able to construct one of the following: DFA *or* NFA *or* ε -NFA *or* regular expression

When is it not?

If we can show that no FA can be built for a language

How to prove languages are ***not*** regular?

What if we cannot come up with any FA?

A) Can it be language that is not regular?

B) Or is it that we tried wrong approaches?

How do we *decisively* prove that a language is not regular?

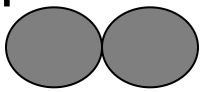
"The hardest thing of all is to find a black cat in a dark room, especially if there is no cat!" *-Confucius*

Example of a non-regular language

Let L = {w | w is of the form 0^n1^n , for all n≥0}

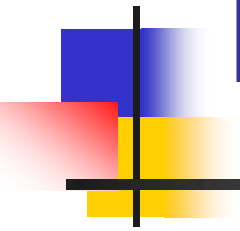
- *Hypothesis: L is not regular*
- Intuitive rationale: How do you keep track of a running count in an FA?
- A more formal rationale:
  - By contradition, if L is regular then there should exist a DFA for L.
  - Let k = number of states in that DFA.
  - Consider the special word w= 0^k1^k => w $\in$ L
  - DFA is in some state p_i, after consuming the first i symbols in w

Rationale…

- Let $\{p_0, p_1, \ldots p_k\}$ be the sequence of states that the DFA should have visited after consuming the first k symbols in w which is 0^k
- But there are only k states in the DFA!
- ==> at least one state should repeat somewhere along the path (by  +  Principle)
- ==> Let the repeating state be $p_i = p_J$ for $i < j$
- ==> We can fool the DFA by inputing $0^{(k-(j-i))}1^k$ and still get it to accept (note: k-(j-i) is at most k-1).
- ==> DFA accepts strings w/ unequal number of 0s and 1s, implying that the DFA is wrong!



The Pumping Lemma for Regular Languages



What it is?

The Pumping Lemma is a property of all regular languages.

How is it used?

A technique that is used to show that a given language is not regular

Pumping Lemma for Regular Languages

Let L be a regular language

Then *there exists* some constant ***N*** such that *for every* string $w \in L$ s.t. $|w| \geq N$, *there exists* a way to break w into three parts, $w=xyz$, such that:

1. $y \neq \varepsilon$
2. $|xy| \leq N$
3. For all $k \geq 0$, all strings of the form $xy^kz \in L$

This property should hold for all regular languages.

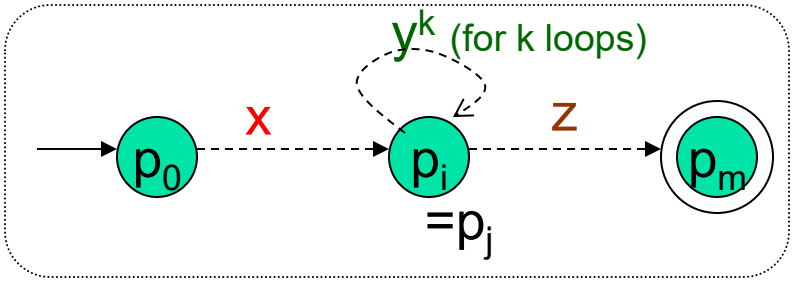
Definition: *N* is called the "Pumping Lemma Constant"

Pumping Lemma: Proof

- L is regular => it should have a DFA.
  - Set *N* := number of states in the DFA
- Any string $w \in L$, s.t. $|w| \geq N$, should have the form: $w=a_1a_2\ldots a_m$, where $m \geq N$
- Let the states traversed after reading the first N symbols be: $\{p_0, p_1, \ldots p_N\}$
  - ==> There are N+1 p-states, while there are only N DFA states
  - ==> at least one state has to repeat
    i.e, $p_i = p_J$ where $0 \leq i < j \leq N$ (by PHP)

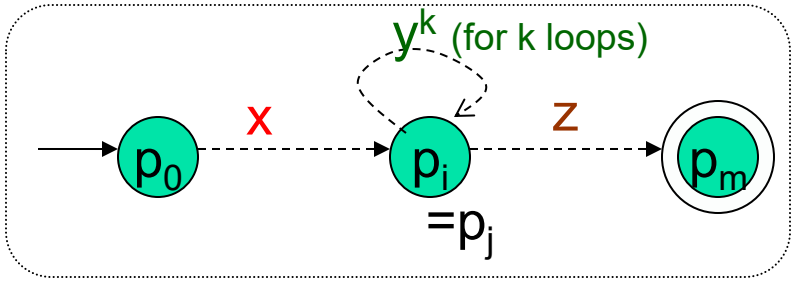
Pumping Lemma: Proof…

- => We should be able to break w=xyz as follows:
  - $x=a_1a_2..a_i$; $y=a_{i+1}a_{i+2}..a_J$; $z=a_{J+1}a_{J+2}..a_m$
  - x's path will be $p_0..p_i$
  - y's path will be $p_i\, p_{i+1}..p_J$ (but $p_i=p_J$ implying a loop)
  - z's path will be $p_Jp_{J+1}..p_m$
- Now consider another string $w_k=xy^kz$, where k≥0
- Case k=0
  - DFA will reach the accept state p_m
- Case k>0
  - DFA will loop for y^k, and finally reach the accept state p_m for z
- In either case, $w_k \in L$ This proves part (3) of the lemma



Pumping Lemma: Proof…

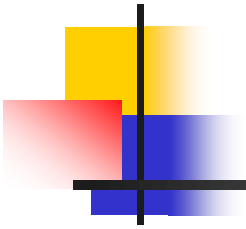
- For part (1):
  - Since i<j, y ≠ ε



- For part (2):
  - By PHP, the repetition of states has to occur within the first N symbols in w
  - ==> |xy|≤N

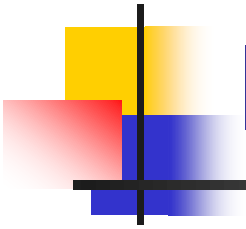
□

The Purpose of the Pumping Lemma for RL



- To prove that some languages *cannot be* regular.

How to use the pumping lemma?



Think of playing a 2 person game

- Role 1: ***We*** claim that the language cannot be regular
- Role 2: An ***adversary*** who claims the language is regular
- We show that the adversary's statement will lead to a contradiction that implies pumping lemma *cannot* hold for the language.
- We win!!

How to use the pumping lemma? (The Steps)

1. (we) L is not regular.
2. (adv.) Claims that L is regular and gives you a value for N as its P/L constant
3. (we) Using N, choose a string $w \in L$ s.t.,
   1. $|w| \geq N$,
   2. Using w as the template, construct other words w_k of the form xy^kz and show that at least one such $w_k \notin L$

      => this implies we have successfully broken the pumping lemma for the language, and hence that the adversary is wrong.

(Note: In this process, we may have to try many values of k, starting with k=0, and then 2, 3, .. so on, until $w_k \notin L$)

Note: We don't have any control over N, except that it is positive.
We also don't have any control over how to split w=xyz,
but xyz should respect the P/L conditions (1) and (2).

Using the Pumping Lemma

- What WE do?

3. Using *N*, we construct our template string *w*
4. Demonstrate to the adversary, either through pumping up or down on *w*, that some string $w_k \notin L$ (this should happen regardless of w=xyz)

- What the Adversary does?

1. Claims L is regular
2. Provides *N*

Note: This N can be anything (need not necessarily be the #states in the DFA. It's the adversary's choice.)

Example of using the Pumping Lemma to prove that a language is not regular

Let L_{eq} = {w | w is a binary string with equal number of 1s and 0s}

- Your Claim: L_{eq} is not regular
- Proof:
  - By contradiction, let L_{eq} be regular → adv.
  - P/L constant should exist → adv.
    - Let *N* = that P/L constant
  - Consider input w = 0^N1^N → you
    (your choice for the template string)
  - By pumping lemma, we should be able to break w=xyz, such that: →you
    1) $y \neq \varepsilon$
    2) $|xy| \leq N$
    3) For all $k \geq 0$, the string xy^kz is also in L

Template string $w = 0^N1^N$ = 00 …. 011 … 1 (N, N)

Proof…

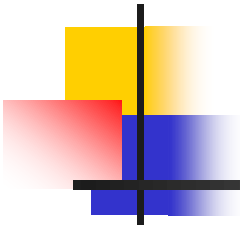
- Because $|xy| \leq N$, xy should contain only 0s
  - (This and because $y \neq \varepsilon$, implies $y=0^+$)
- Therefore x can contain *at most* N-1 0s
- Also, all the N 1s must be inside z
- By (3), any string of the form $xy^kz \in L_{eq}$ for all $k \geq 0$
- Case k=0: xz has at most N-1 0s but has N 1s
- Therefore, $xy^0z \notin L_{eq}$
- This violates the P/L (a contradiction)

➔ you

Setting k=0 is referred to as **"pumping down"**

Setting k>1 is referred to as **"pumping up"**

Another way of proving this will be to show that if the #0s is arbitrarily pumped up (e.g., k=2), then the #0s will become exceed the #1s



Exercise 2

Prove L = {$0^n 1 0^n$ | n≥ 1} is not regular

Note: This n is not to be confused with the pumping lemma constant N. That *can* be different.

In other words, the above question is same as proving:

- L = {$0^m 1 0^m$ | m≥ 1} is not regular

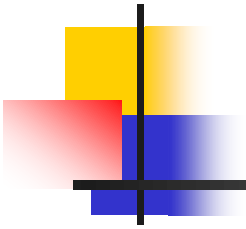
Example 3: Pumping Lemma

Claim: L = { 0^i | i is a perfect square} is not regular

- Proof:
  - By contradiction, let L be regular.
  - P/L should apply
  - Let *N* = P/L constant
  - Choose w=0^{N^2}
  - By pumping lemma, w=xyz satisfying all three rules
  - By rules (1) & (2), y has between 1 and N 0s
  - By rule (3), any string of the form xy^kz is also in L for all k≥0
  - Case k=0:
    - #zeros (xy^0z) = #zeros (xyz) - #zeros (y)
    - $N^2 - N \leq$ #zeros (xy^0z) $\leq N^2 - 1$
    - $\mathbf{(N-1)^2} < N^2 - N \leq$ #zeros (xy^0z) $\leq N^2 - 1 < \mathbf{N^2}$
    - $xy^0z \notin L$
    - But the above will complete the proof ONLY IF N>1.
    - … (proof contd.. Next slide)

Example 3: Pumping Lemma

- (proof contd…)
  - If the adversary pick N=1, then $\mathbf{(N-1)^2} \leq N^2 - N$, and therefore the #zeros(xy^0z) could end up being a perfect square!
  - This means that pumping down (i.e., setting k=0) is not giving us the proof!
  - So lets try pumping up next…
- Case k=2:
  - #zeros (xy^2z) = #zeros (xyz) + #zeros (y)
  - $N^2 + 1 \leq$ #zeros (xy^2z) $\leq N^2 + N$
  - $\mathbf{N^2} < N^2 + 1 \leq$ #zeros (xy^2z) $\leq N^2 + N < \mathbf{(N+1)^2}$
  - $xy^2z \notin L$ 
  - (Notice that the above should hold for all possible N values of N>0. Therefore, this completes the proof.)



Exercises

- Prove that $L = \{ww \mid w \in \{a, b\}^*\}$is not regular?
- Prove that $L = \{a^p \mid p \textit{ is prime}\}$is not regular?